

Lei Zhang

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Research Interests

- *Multimodal Intelligence* — Advancing both the native capabilities and agentic wisdom of multimodal models to empower them with more sophisticated reasoning abilities in understanding and generation.
- *Scalable Training Methodology* — Developing data curation, evaluation, and training strategies to scale multimodal models toward stronger and more generalizable intelligence.

Experiences

- 2026 – Now Research Intern, Nvidia Research
 Host: Hanrong Ye, Pavlo Molchanov
- 2025 Research Scientist Intern, Meta GenAI
 Host: Zecheng He, Junjiao Tian, Zhipeng Fan

Education

- 2024 – Now Ph.D., Computer Science, University of California San Diego
 Advisor: Prof. Julian McAuley
- 2024 M.S., Computer Science, Zhejiang University, China
- 2021 B.S., Software Engineering, South China University of Technology, China

Awards & Honors

- 2025 Notable Reviewer Award, ICLR 2025
- 2023 Xiaomi Scholarship (10,000 RMB)

Publications

* → equal contribution † → core contributor

[Google Scholar Profile](#)

Preprints

- [1] Wei Zhou, Xiongwei Zhu, Lingdong Kong, Bo Chen, **Lei Zhang**, Yongyuan Liang, Xiaoxia Hou, Ye Tian, Xian Sun, Yingshuo Wang, Linfeng Li, Shengqiong Wu, Leigang Qu, Feng Li, Wei Liu, Julian McAuley, Tat-Seng Chua.
On-Policy Self-Distillation in Diffusion Models (2026). [PDF](#)
Featured by **ByteDance Seed**

- [2] Fangxun Shu, **Lei Zhang**^{*}, Yongjie Ye, Yue Liao, Haiyang Guo, Gengyuan Liu, Liang Xiao, Shuicheng Yan, Chao Feng.
Variance-aware Sequence Policy Optimization (2026).

Refereed Papers

- [1] **Lei Zhang**, Junjiao Tian, Zhipeng Fan, Kunpeng Li, Jialiang Wang, Weifeng Chen, Markos Georgopoulos, Felix Juefei-Xu, Manling Li, Julian McAuley, Zecheng He.
Think in Strokes, Not Pixels: Process-driven Image Generation via Interleaved Reasoning
European Conference on Computer Vision (ECCV 2026). [PDF](#)
#1 Huggingface Daily Paper
Featured by DeepLearning.AI in *The Batch* and *Meta AI*
- [2] Haoxiang Zhang[†], Qixin Xu[†], Zhuofeng Li[†], **Lei Zhang**[†], Pengcheng Jiang, Caiming Xiong, Yu Zhang, Julian McAuley.
Masking Stale Observations Helps Search Agents - Until It Doesn't: A Regime Map and Its Mechanism. 2026.
Empirical Methods in Natural Language Processing (EMNLP 2026). [PDF](#)
- [3] Fangxun Shu[†], Yue Liao[†], **Lei Zhang**[†], Le Zhuo, Chenning Xu, Guanghao Zhang, Haonan Shi, Weilong Dai, Tao Zhong, Zhelun Yu, Wanggui He, Siming Fu, Haoyuan Li, Hongsheng Li, Hao Jiang.
LLaVA-MOD: Making LLaVA Tiny Via MOE-Knowledge Distillation
International Conference on Learning Representations (ICLR 2025). [PDF](#)
- [4] Sucheng Ren, Xianhang Li, Haoqin Tu, Feng Wang, Fangxun Shu, **Lei Zhang**, Jieru Mei, Linjie Yang, Peng Wang, Heng Wang, Alan Yuille, Cihang Xie.
Autoregressive pretraining with mamba in vision
International Conference on Learning Representations (ICLR 2025). [PDF](#)
- [5] Wanggui He, Siming Fu, Mushui Liu, Xierui Wang, Wenyi Xiao, Fangxun Shu, Yi Wang, **Lei Zhang**, Zhelun Yu, Haoyuan Li, Ziwei Huang, Leilei Gan, Hao Jiang.
MARS: Mixture of Auto-Regressive Models for Fine-grained Text-to-image Synthesis
Association for the Advancement of Artificial Intelligence (AAAI 2025). [PDF](#)
- [6] Jiaqi Chen, Xiaoye Zhu, Tianyang Liu, Ying Chen, Xinhui Chen, Yiwen Yuan, Chak Tou Leong, Zuchao Li, Tang Long, **Lei Zhang**, Chenyu Yan, Guanghao Mei, Jie Zhang, Lefei Zhang.
Imitate Before Detect: Aligning Machine Stylistic Preference for Machine-Revised Text Detection
Association for the Advancement of Artificial Intelligence (AAAI 2025). [PDF](#)
Oral (4.6%)
- [7] **Lei Zhang**, Jie Zhang, Bowen Lei, Subhabrata Mukherjee, Xiang Pan, Bo Zhao, Caiwen Ding, Yao Li, Dongkuan Xu.
Accelerating Dataset Distillation via Model Augmentation
IEEE / CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023). [PDF](#)
Highlight (2.5%)
- [8] **Lei Zhang**, Zhibo Wang, Xiaowei Dong, Yunhe Feng, Xiaoyi Pang, Zhifei Zhang, Kui Ren.
Towards Fairness-aware Adversarial Network Pruning

IEEE International Conference on Computer Vision (**ICCV 2023**). [PDF](#)
Featured by Adobe Research

- [9] Jie Zhang*, **Lei Zhang***, Gang Li, Chao Wu.
Adversarial Examples for Good: Adversarial Examples Guided Imbalanced Learning
 IEEE International Conference on Image Processing (**ICIP 2022**). [PDF](#)
- [10] Jie Zhang, Bo Li, Jianghe Xu, Shuang Wu, Shouhong Ding, **Lei Zhang**, Chao Wu.
Towards Efficient Data Free Black-box Adversarial Attack
 IEEE / CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2022**).
[PDF](#)

Technical Reports

- [1] **Lei Zhang**, Fangxun Shu, Tianyang Liu, Sucheng Ren, Hao Jiang, Cihang Xie.
Filter & Align: Leveraging Human Knowledge to Curate Image-Text Data. 2024. [PDF](#)
- [2] Fangxun Shu*, **Lei Zhang***, Hao Jiang, Cihang Xie.
Audio-visual LLM for Video Understanding. 2024. [PDF](#)
- [3] Wenqiao Zhang, Tianwei Lin, Jiang Liu, Fangxun Shu, Haoyuan Li, **Lei Zhang**, Wanggui He, Hao Zhou, Zheqi Lv, Hao Jiang, Juncheng Li, Siliang Tang, Yueting Zhuang.
HyperLLaVA: Dynamic Visual and Language Expert Tuning for Multimodal Large Language Models. 2024. [PDF](#)

Professional Activities & Service

Reviewer, CVPR, NeurIPS, ICLR, ECCV, ACL

Notable reviewer, ICLR 2025

Teaching

Spring 2026 **Teaching Assistant**, Web Mining and Recommender Systems, UCSD
 Fall 2025 **Teaching Assistant**, Machine Learning for Music, UCSD